

Shiv Nadar University Chennai
CS3807 – Deep Learning Laboratory
Experiment 2
Implementation of Convolutional Neural Network (CNN) for CIFAR-10 Image Classification using TensorFlow/Keras

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

Implementation of Convolutional Neural Network (CNN) for CIFAR-10 Image Classification using TensorFlow/Keras.

2. Dataset

Dataset: CIFAR-10

Training Images: 50,000

Testing Images: 10,000

Classes: 10

Image Size: 32×32 (RGB)

3. Image Preprocessing

CIFAR-10 images are stored as $32 \times 32 \times 3$ matrices representing RGB channels.

Perform the following preprocessing:

1. Load the CIFAR-10 dataset.
2. Display sample images.
3. Normalize pixel values to the range $[0,1]$.
4. Convert categorical class labels into one-hot encoded vectors.

4. Experimental Procedure

Task 1: Dataset Exploration

- Load the CIFAR-10 dataset.
- Display sample images from different classes.
- Plot the class distribution to verify the dataset balance.

Expected Output: Dataset summary and sample images.

Task 2: Data Preprocessing

- Normalize pixel values.
- One-hot encode the categorical labels.

Task 3: Model Construction

Construct a CNN architecture with:

- Conv2D layers for feature extraction
- MaxPooling2D layers for downsampling
- Flatten layer

- Dense and Dropout layers for classification
- Display `model.summary()`.

Task 4: Model Training

Compile using

- Optimizer: Adam
- Loss: Categorical Crossentropy
- Metric: Accuracy

Train the model and monitor the validation metrics.

Task 5: Model Evaluation

Compute and analyze

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report
- Feature Map Visualization
- Pooling Comparison
- Trainable Parameter Analysis

5. Mandatory Plots



Figure 1: CIFAR-10 Sample Images

Inference: The images show sample objects from the CIFAR-10 dataset. They are loaded correctly in RGB format and the 32x32 resolution is visible.

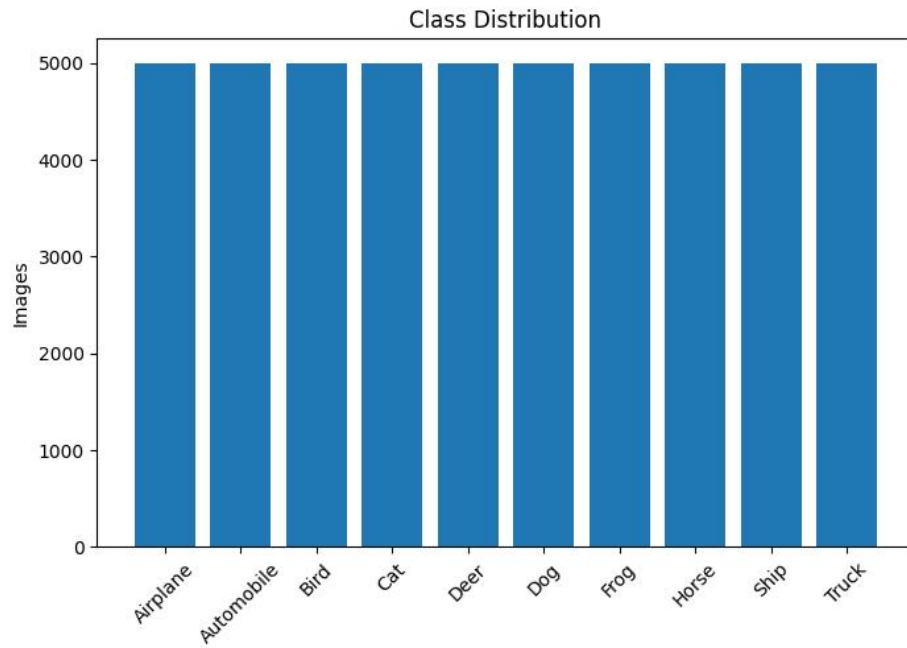


Figure 2: Class Distribution

Inference: The bar chart shows the distribution of 10 CIFAR-10 categories. This means that the dataset is balanced, so the model won't favor one class over another during training.

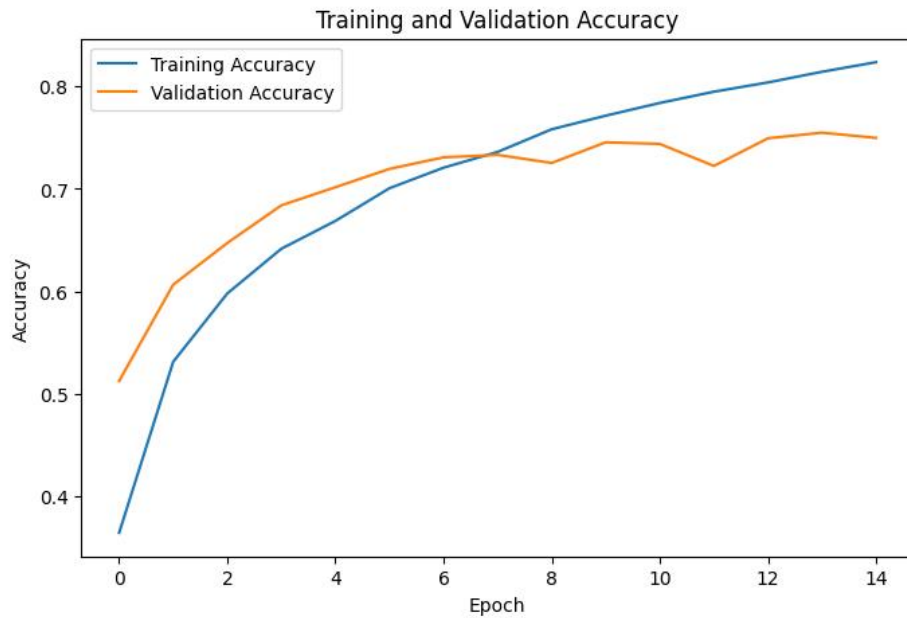


Figure 3: Training and Validation Accuracy vs Epoch

Inference: Both training and validation accuracy increase over the epochs. This means that the model is learning well and generalizing to the new data.

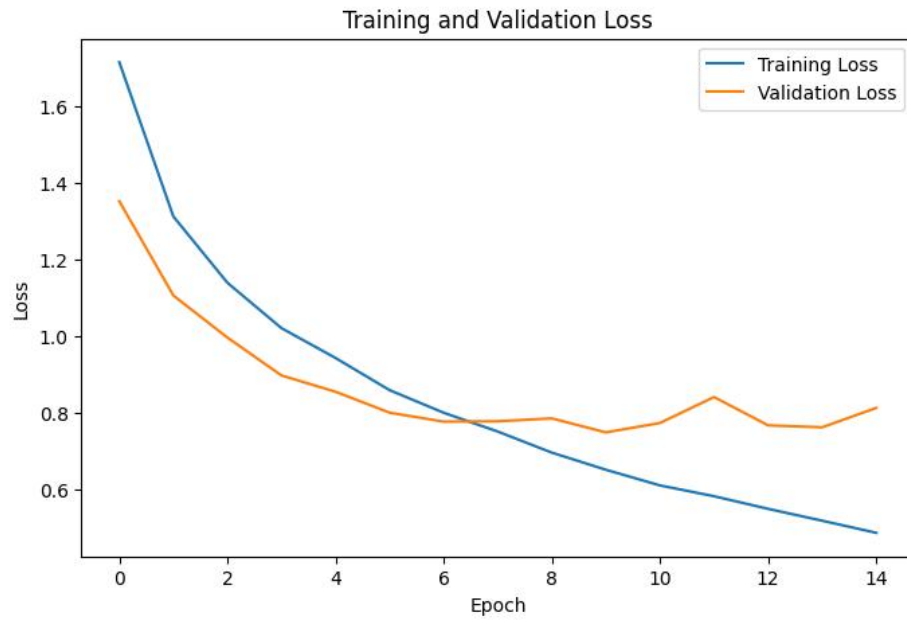


Figure 4: Training and Validation Loss vs Epoch

Inference: The loss curves for both training and validation show convergence over time. This indicates that the Adam optimizer is minimizing the categorical crossentropy loss effectively.

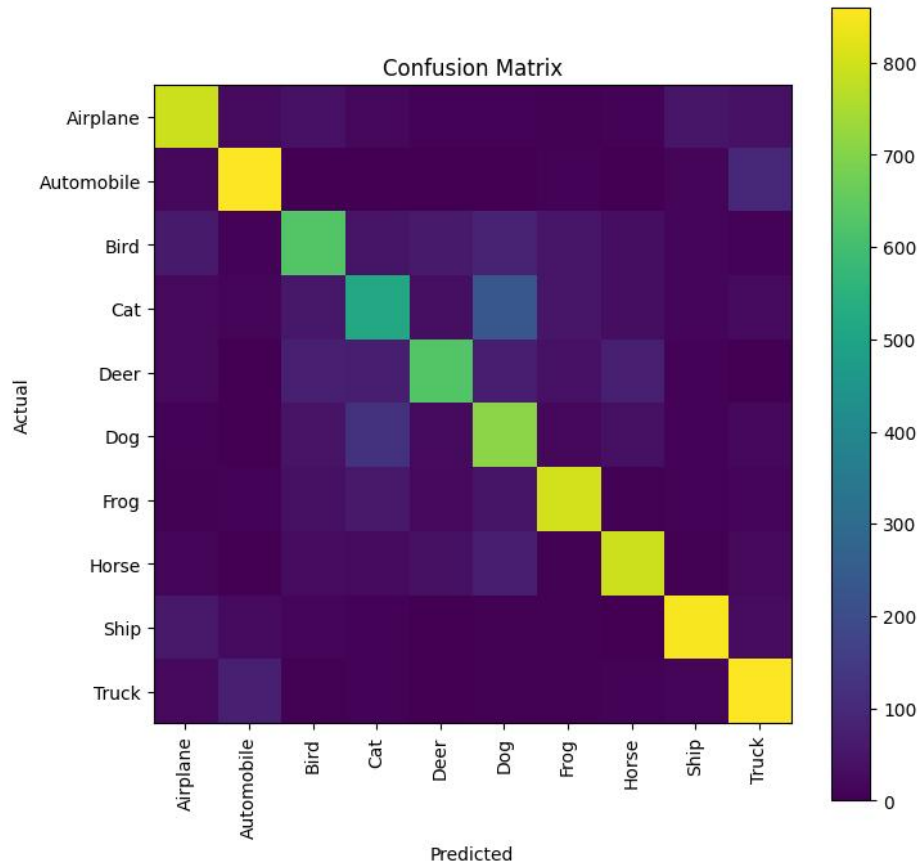


Figure 5: Confusion Matrix

Inference: The diagonal has high values which means that most of the predictions are correct. Some misclassifications occur between visually similar categories.

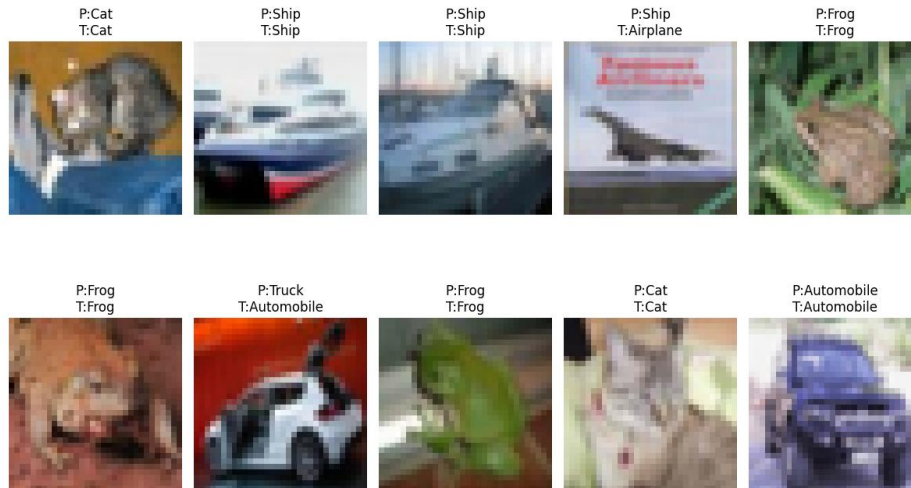


Figure 6: Prediction Samples

Inference: The sample predictions show what the model predicted next to the actual labels. Most predictions are correct. This shows that the model can recognise well.

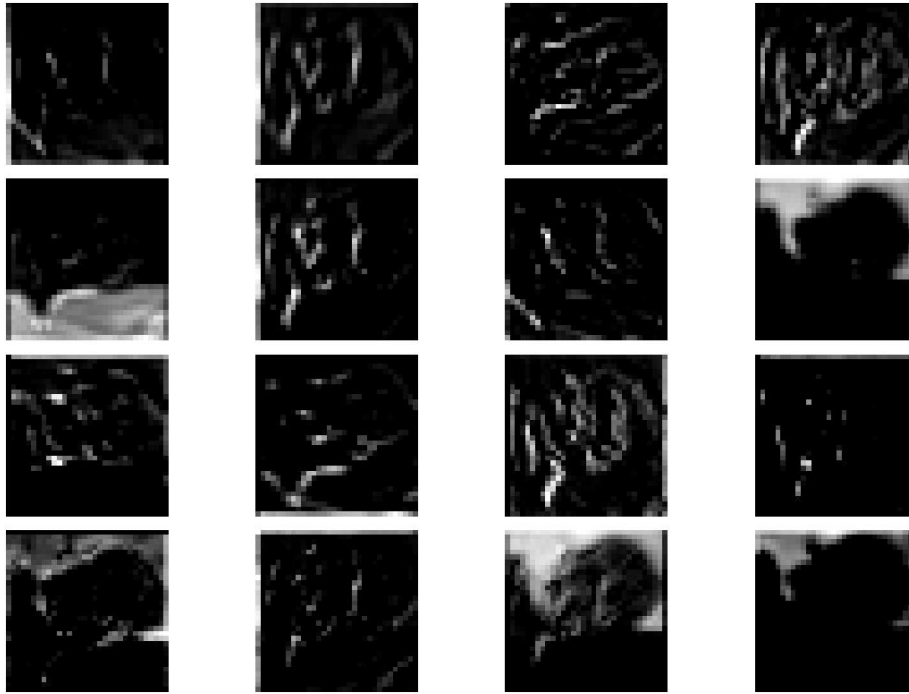


Figure 7: Feature Map Visualization

Inference:The feature maps show what the convolutional layers are detecting.The early layers detect simple patterns like edges and curves.The deeper layers detect more complex patterns like shapes and objects.

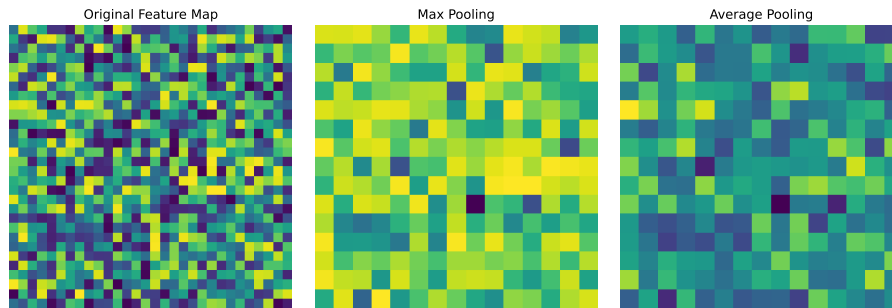


Figure 8: Max Pooling vs Average Pooling Comparison

Inference:Max pooling keeps the strongest features and edges while average pooling smooths things.Here,max pooling worked better since it preserves important patterns.

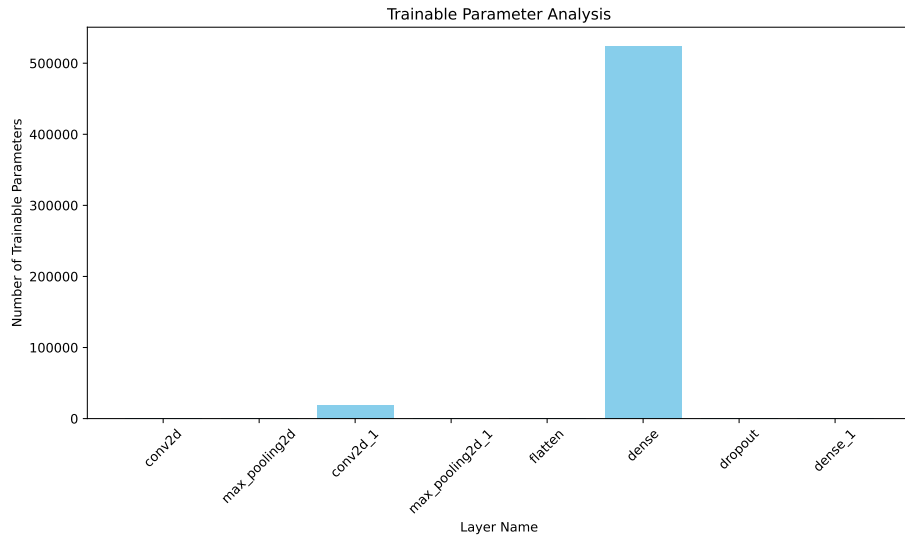


Figure 9: Trainable Parameter Analysis

Inference: The analysis shows most of the trainable parameters are in the dense layers. The convolutional layers have fewer parameters compared to dense layers.

6. Results

Table 1: Model Evaluation Metrics

Metric	Value
Test Accuracy	74.43%
Test Loss	0.8288
Precision	0.7462
Recall	0.7443
F1 Score	0.7433

Source Files

- **Google Colab Notebook:** Colab Notebook
- **GitHub Repository:** GitHub Repository

7. References

1. Goodfellow et al., *Deep Learning*.
2. CIFAR-10 Dataset Documentation.
3. TensorFlow/Keras Documentation.